

Diagnosis of Leukemia Disease through Deep Learning using Microscopic Images

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Abstract— The abnormal production of WBC (white blood cell) in the bone marrow is known as leukemia. Leukemia is one of the most affecting diseases around the globe. Several types of ML (machine learning) and DL (deep learning) classification models have been presented in the literature to detect this disease, but they still possess some drawbacks. This study proposes a framework for detecting five classes of leukemia: ALL, AML, CLL, CML, and normal cell. In this research, a pre-trained DCNN (deep convolutional neural network) has been proposed for the detection of leukemia through microscopic images. Pre-processing of microscopic images improves the contrast and removes the noise by enhancing and filtering images. Segmentation of microscopic images is used to highlight the area of the disease. Alex-Net and ResNet-34 architecture are used for classification purposes. After comparing these two models through statistical parameters, ResNet-34 attained the most accurate result than Alex-Net using the publicly available ALL-IDB dataset, the evaluation through the statistical parameters revealed that ResNet-34 attained a classification accuracy of 98.4% on ALL, 98.4% on AML, 98.13% over CLL, 98.14 over CML. AlexNet attained 96.1% classification accuracy on ALL, 95.5% on AML, 95.7% on CLL, and 96.8% on CML. The proposed framework significantly outperforms existing technologies and can be used in clinical applications.

Keywords— Artificial intelligence, leukemia detection, white blood cells, bone marrow, segmentation.

I. INTRODUCTION

Leukemias are blood malignancies that occur due to the neoplastic proliferation of bone marrow precursor cells resulting in impaired immunity, bleeding tendencies, and ineffective erythropoiesis [1]. Every year, the number of new instances of leukemia increases, accompanied by a high death rate, which can lead to a delay in leukemia treatment and diagnosis [2]. Figure 1 shows the normal and leukemia-affected images.

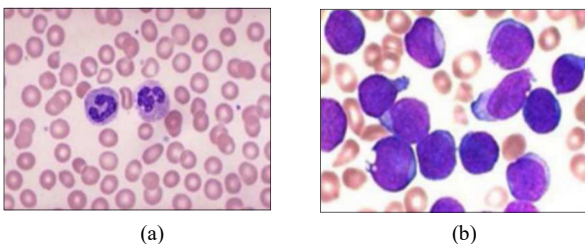


Fig. 1. Microscopic Images (a) Normal (b) Leukemia [2]

Acute leukemia and chronic leukemia are two types of leukemia. Both above-mentioned types have two subtypes: myeloid and lymphocytic. Myeloid Leukemia is subdivided into AML (Acute myelogenous Leukemia) and CML (Chronic Myeloid Leukemia). Similarly, lymphocytic leukemia is also subdivided into CLL (Chronic Lymphoid Leukemia), and ALL (Acute Lymphoblastic Leukemia) depict in table 1.

TABLE I. TYPES OF LEUKEMIA

Types	Lymphoid	Myeloid
AL (Acute Leukemia)	ALL (Acute Lymphoid Leukemia)	AML (Acute Myeloid Leukemia)
CL (Chronic Leukemia)	CLL (Chronic Lymphoid leukemia)	CML (Chronic Myeloid Leukemia)

A. Acute Lymphoid Leukemia (ALL)

This form of malignancy is most common in children and develops due to a lack of development or a negative influence on the bone marrow. When WBCs grow fast, the bone marrow's ability to function is severely harmed, resulting in cancer [3].

B. Acute Myeloid Leukemia (AML)

This form of malignancy develops due to a lack of development or a negative influence on the bone marrow. When WBCs grow fast, the bone marrow's ability to function is severely harmed, resulting in cancer [4].

C. Chronic Lymphocytic Leukemia (CLL)

It affects senior citizens who are suffering from old age disorders. These lymphoid are harmed at the early stage. There are no signs or symptoms [5].

D. Chronic Myeloid Leukemia (CML)

It affects patients between the ages of 35 and 45. Myeloid cells undergo genetic alterations at an early stage [6].

A critical and time-consuming stage in the identification of leukemia is the evaluation of optical blood smears under the supervision of an expert. To address these issues, several decision support system methods for the analysis of microscopic images based on ML and DL have been developed. However, there are several flaws in these approaches that need to be overcome in terms of efficiency, learning process, and accuracy. When compared to manual

methods, the microscopic technique was found to be cost-effective, more efficient, quick, and accurate [7].

Microscopic blood images are the most important method for the detection of leukemia disease. The microscope technique is not the only one for the detection of leukemia. There is Another Way for diagnosing leukemia is interventional radiology. However, biopsy, Percutaneous, catheter drainage, aspiration are examples of radiological methods that suffer from the sensitivity and resolution limitations of the imaging modality.[8] In addition, other techniques for identifying leukemia types, such as aCGH (Array-based Comparative Genomic Hybridization), LDI-PCR (Long Distance Inverse Polymerase Chain Reaction), Molecular Cytogenetics, aCGH (Array-based Comparative Genomic Hybridization, and required a lot of effort and time [9]. Due to the limitations of time and cost, microscopic images tests and bone marrow are commonly used to identify leukemia subtypes.

Artificial intelligence (AI) technologies are quickly becoming integrated into our daily lives. Furthermore, several efforts are being made in the fields of medical science and healthcare to include Artificial Intelligence developments for real-time medical diagnosis and treatment [10-12].

The detection and diagnosis within the medical domain such as hypertensive retinopathy [13, 14], Papilledema [15, 16], Brain Tumor [17-19], Glaucoma [20-22], Melanoma [23, 24], Alzheimer's [25] and Central Serous Retinopathy [26, 27] can be performed through machine learning and deep learning methodologies.

The remainder of the paper is laid out as follows: The literature review of previous studies is presented in Section II. In Section III, the proposed methodology for diagnosing leukemia is described. The proposed method's results and discussion are presented in Section IV. The conclusion is found in Section V.

II. LITERATURE REVIEW

Sarmad et al. [28] Presented a method for detecting ALL through microscopic images and ALL-IDB dataset used, which achieved 93% accuracy and 92.1% sensitivity, and 91.2% specificity. Rahman et al. [29] Developed a technique for the diagnosis of acute lymphoid leukemia (ALL) by using the morphological feature, color, and textural feature through blood microscopic smear images. ALL-IDB dataset was used, which attained an accuracy of 93%. Aftab et al. [30] BigDL Apache spark publicly available library from the micrographs of leukemia using Google-Net deep transfer learning, and CNN was proposed. This had a 97.33% accuracy rate. Kumar Das et al. [31] Introduced a hybridizing MobilenetV2 and ResNet18 deep CNN architecture for the detection of ALL. Two publicly available ALL-IDB1 and ALL-IDB2 datasets were employed, with an accuracy of 97.92% and 96%, respectively. Pansombut et al. [32] Discovered a technique for grouping ALL and its subdirectories using the Convolutional Neural Network. Two standers random forest, multilayer perceptron (MLP) machine learning classifiers were used for equivalence checking. A dataset (ALL-IDB, ASH) of 364 images was used for analysis. As a result, 80 % accuracy was obtained. Pawan et al. [33] The author proposed a system for the diagnosis of ALL using

Naive Bayes and KNN Classifier. 100 images were used to test the classifier, which gives 92.8% accuracy. Thanh et al. [34] Put forwarded a CNN for the prediction of healthy and leukemia-affected images. ALL-IDB1 dataset was used to achieve an accuracy of 96.60%, respectively. Agaian et al. [35] Proposed the system, which displayed a technique for categorizing multi-nucleate Blood images. For the classification, the author was used SVM classifier. This system attained an accuracy of 94 % by using 98 images from ALL-IDB.

TABLE II. COMPARATIVE ANALYSIS OF MACHINE AND DEEP LEARNING TECHNIQUE

Ref. #	Year	Authors	Datasets and images	Results
[28]	2019	Sarmad Shafique et al.	ALL-IDB 1 108	Accuracy of 93.7%, Sensitivity 92%, Specificity of 91%
[29]	2018	Ashikur Rahman et al.	ALL-IDB 108	Accuracy 93%
[30]	2021	Muhammad Omer Aftab et al.	Local dataset 1200	Accuracy 97.33%
[31]	2021	Pradeep Kumar Das et al.	ALL-IDB1 108 ALL-IDB2 260	Accuracy 97.92 % Accuracy 96.00 %
[32]	2019	Tatdow Pansombut et al.	ALL-IDB and ASH 364	Accuracy 80%
[33]	2018	Sachin Paswan et al.	Local dataset 100	Accuracy 92.8%
[34]	2018	T. T. P. Thanh et al.	ALL-IDB1 108	Accuracy 96.6%
[35]	2016	Sos Agaian et al.	ALL-IDB1 98	Accuracy 94%

III. PROPOSED METHODOLOGY

The publicly available database ALL-IDB has been expanded using the augmentation technique to enlarge the number of images required for training deep CNN (convoluted neural networks). However, segmentation is used to highlight the affected area of the disease. We used pre-trained architecture AlexNet and ResNet-34 for classification purposes. Based on statistical evaluation of the parameters, Multiple Convolutional Neural Network architectures were compared in terms of performance.

A. Dataset

A publicly available ALL-IDB dataset is used that has been imported from [36] in this research. This dataset is available in two versions: ALL-IDB1 and ALL-IDB2. This dataset contains images taken with a PowerShot G5 camera with a native resolution of 2592 x 1994. The dataset of ALL-IDB comprised 108 images, forty-nine images were from leukemia-affected images and fifty-nine images were from healthy/normal images. Specialists labeled the lymphoblasts, obtaining 510 elements from 370. The ALL-IDB 2 data set included 260 single-cell images, 130 of which were from patients with leukemia and 130 of which were normal images [37]. Fig. 2 shows the framework of the proposed method.

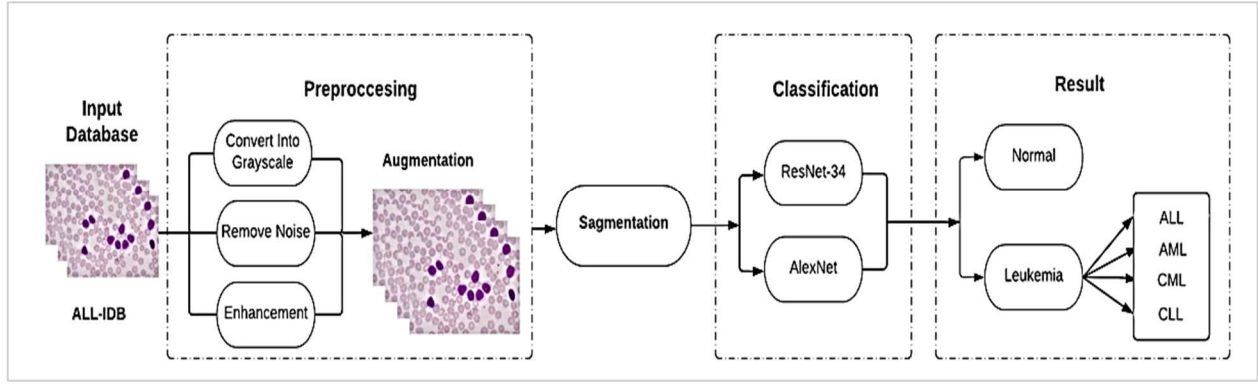


Fig. 2. Workflow diagram of the proposed methodology

B. Pre-processing

In preprocessing of the microscopic image, RGB images were converted into grayscale images. After the images are converted into grayscale, the microscopic image is filtered with a 3×3 size median filter to remove noise and improve image quality. The median filter is a non-linear digital filter that is commonly used to remove noise from images or signals. This is due to the images in the ALL-IDB dataset having a lot of noise and low contrast. The network can now extract significant information from tiny pictures since the noise has been reduced. To identify edges, a high pass filter is applied to the obtained microscopic image. The enhanced microscopic image is then created by merging the edge-identified microscopic image with the original image. Fig. 3 shows the preprocessing phases.



Fig. 3. Preprocessing phases for microscopic images

C. Data Augmentation:

To increase the number of images in the dataset, data augmentation techniques are frequently used in the DL method. If the original images as well as their sub-versions were used to train ML and DL models, they would be more generalizable. The sample datasets for deep learning techniques are very limited, with only 368 samples in the original ALL-IDB dataset. To maximize the dataset, data augmentation techniques such as rotations, zooming, width shift, height shift, horizontal flip, and hearing are used. As a result, the sample size in the dataset is increased after applying image augmentation methods. In the ALL-IDB datasets, the sample size increased to 3959 after data augmentation.

D. Segmentation

Semantic segmentation is the method of combining portions of images that belong to the same category. DeepLabv3+, one version of DeepLab, is used for segmentation tasks in the proposed work. For feature extraction, DeepLabv3+ uses ResNet or MobileNet v2 as the pre-trained CNN network. In the encoder section, it uses atrous separable convolution in atrous spatial pyramid pooling to extract information from the input image. Spatial and pointwise convolution are examples of separable convolution. To produce the result, it first executes

independent spatial convolution on each channel, followed by a pointwise convolution. The decoder module is responsible for fine-tuning the segmentation result.

E. Classification

After the preprocessing and the segmentation of microscopic images, we will do the classification. To achieve the best accuracy, the performance of several pre-trained deep network architectures was compared. The architecture of Alex-Net, ResNet-34 has been modified using deep learning in accordance with the isolated classes included in the microscopic image classification. The selection of these classifiers was based on their excellent results in classification and prediction problems.

1) ResNet-34

ResNet-34 is a 34-layer model that has been pre-trained. Deep neural networks performance is influenced by their architecture and dataset. Better results are obtained by using a large dataset in a deep network of CNN architecture. Furthermore, after a certain depth, performance degrades as the network becomes deeper. ResNet-34 is made up of 34 layers, one of which is a convolution layer and pooling layer, as well as four other layers which have the same sequence. Each layer is interspersed with feature maps of sizes 64, 128, 256, and 512, with 3×3 convolutions [38].

2) Alex-Net

Alex-Net is the most widely used CNN architecture for reorganization and classification. Alex-Net has a total of eight layers, with three fully connected layers and five convolutional layers. Before the first, second, and fifth convolutional layers, the Max Pooling layers were combined. The third and fourth convolutional layers in this architecture are directly connected. Each of the eight layers uses the ReLU activation function. To assign values for categorization in each class, the SoftMax function is used in the final layer. There is a huge amount of information about Alex-Net architecture in [39].

IV. RESULTS AND DISCUSSION

Several studies have been carried out to assess the performance of CNN for the detection of leukemia. This experimental result aids in determining which systems have achieved the best performance in terms of clinical application and network competence reliability.

The proposed methodology used the ALL-IDB dataset. The dimensions of the input images in Alex-Net Deep CNN have been kept at 227x277. In contrast, ResNet-34 images have a resolution of 224x244 pixels. We have augmented the original dataset to 3959 pictures in this paper to enhance the accuracy of our proposed method using a large dataset by applying augmentation techniques, such as histogram equalization, blurring, translation, rotation, zooming, shearing, and reflection. After applying the augmentation step, the dataset contains 3959 images divided into five classes: CLL, CML, ALL, AML, and normal, 30% of images are used for testing and validation while 70% of the images are used as training data. The training parameters used in the training of the two networks are shown in table III.

TABLE III. TRAINING PARAMETERS OF TWO NETWORKS

Parameters	ResNet-34	Alex-Net
Learning rate	0.0001	0.0001
Max iterations	6000	6000
Epoch	11	11
Loss Function	0.503	0.57764
Frequency	28	28
Elapsed time	24 min 19 sec	26 min

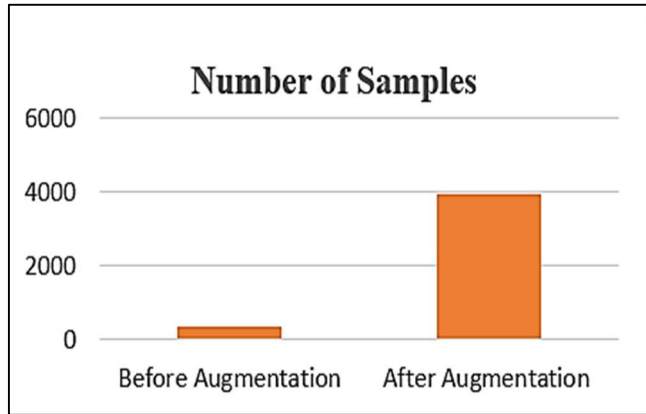


Fig. 4. Number of images before and after augmentation.

A. Evaluation parameters

Accuracy, precision, recall, and F1 score are the performance criteria that are used to evaluate the proposed methodology.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (2)$$

$$\text{Accuracy} = \frac{TP+TN}{(TP+FN+TN+FP)} \times 100 \quad (3)$$

$$\text{F1 - score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100 \quad (4)$$

Table IV shows the comparison between results of Resnet-34 and Alex-net.

TABLE IV. COMPARISON OF RESULTS BETWEEN RESNET-34 AND ALEX-NET

Parameters	ResNet-34				
	ALL	AML	CLL	CML	Healthy
Accuracy	98.4%	98.4%	98.13%	98.14%	99%
F1 score	99%	99.1%	99.24%	99.1%	99.24%
Precision	98%	98.34 %	98.3%	98.3%	98.34%
Recall	99%	98.4%	98%	98.9%	99. %
	Alex-Net				
	ALL	AML	CLL	CML	Healthy
Accuracy	96.1%	95.5%	95.7%	96.8%	96.8%
F1 score	97.9%	96.9%	97.34%	97.1%	97.9%
Precision	95.3%	95.63 %	95%	96%	96%
Recall	95.5%	95.39 %	95.9%	96.3%	96.3%

The Alex-Net and ResNet-34 are used for leukemia detection and classification. For the healthy and ALL cases, the accuracy of ResNet-34 is 98.4%. While F1 score 99%, recall 99%, precision 98%. ResNet-34 has a prediction accuracy of 98.4%, precision of 98.34%, recall of 98.4%, and F1 score of 99.1% for AML. For the prediction of CLL, ResNet-34 gives the accuracy is 98.13%, recall 98%. precision value 98.3%, F1 score are 99.1%. CML give the F1 score is 99.24%, accuracy is 98.14%, recall is 98.9%, precision is 98.34%. However, Whereas, Alex-Net gives the accuracy rate of 96.1% and F1-score 97.9%, precision 95.3%, Recall 95.5% for ALL and healthy. In the case of AML recall 95.39%, accuracy 95.5%, F1 score 96.9%, precision 95.63%. For CLL, prediction accuracy 95.7%, recall 95.9%, F1 score 97.1%, precision value 95%. For the CML, Alex-Net gives the accuracy value of 96.8%, F1 score 97.1%, precision 96%, and Recall 96.3%.

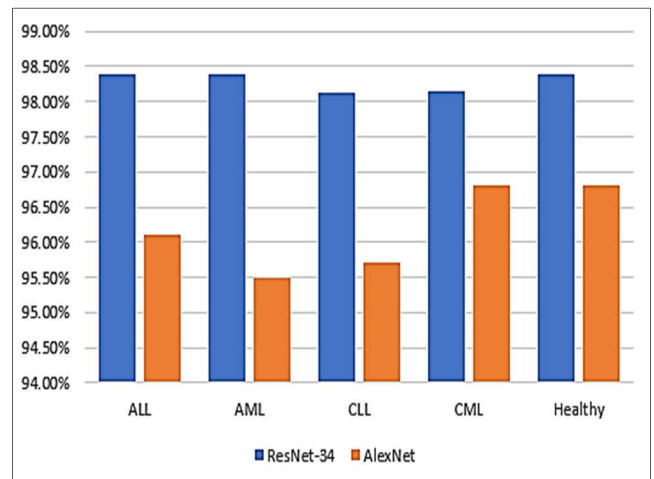


Fig. 5. Accuracy comparison between the ResNet-34 and Alex-Net

Table V shows the proposed model's performance based on the evaluation parameters.

TABLE V. COMPARISON OF PROPOSED METHOD WITH STATE-OF-ART METHODS

Ref. #	Classifiers	Dataset	Accuracy	Recall	Precision
[30]	Google Net	Local dataset	97.33%		
[31]	ResNet-18, MobailnetV 2	ALL-IDB	97.92 % 96.00 %		
[32]	CNN	ALL-IDB ASH	80%		
[34]	CNN	ALL-IDB	96.6%		
Proposed Method	ResNet-34	ALL-IDB	98.4%	99%	98.34%
	Alex-Net	ALL-IDB	96.8%	96.3%	96%

It is concluded that the ResNet-34 achieved excellent results as compared to the Alex-Net classifier. ResNet-34 attained a precision of 98.34%, recall of 99%, and accuracy of 98.4%. Furthermore, all of the evaluation parameters show the robust performance for both classifiers that is above 90%, which represents the efficiency of DL-based algorithms for the prediction and classification of leukemia using microscopic images. However, For validation purposes, previous researches proposed methodologies that used local and small datasets with low accuracy.

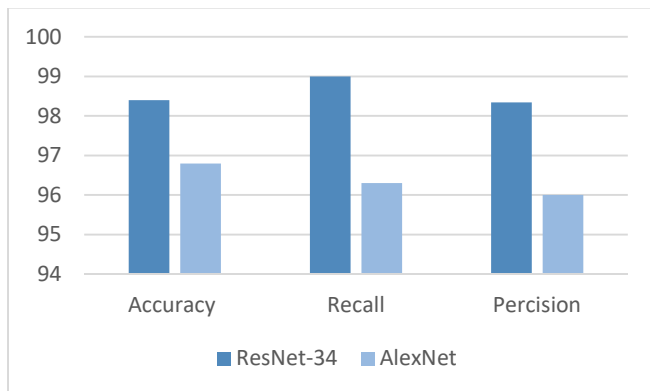


Fig. 6. Performance analysis of proposed methodology.

The relevant work describes the numerous microscopic images for the detection of leukemia using segmentation and classification algorithms. Using publically available datasets ALL-IDB, we have compared the results generated by the proposed model with those acquired by state-of-the-art approaches to verify the performance and resilience of our method. The results show that segmentation and classification improve the performance of our developed framework more than any other state-of-the-art method.

The limitations of previous research have shown that using small datasets produces the best results, but using a large dataset produces inaccurate results. However, other issues arise as a result of using large datasets, such as image resolution, image brightness, color fidelity, and the incapability to correctly detect abnormal cells with high performance. Another significant flaw in deep learning methods is that they were trained and tested on private datasets that are not publicly available.

The main contributions of our research are using publicly available datasets to increase the number of images using the augmentation technique. To the best of our knowledge, no such study is available in literature for detection of leukemia which uses the segmentation as well as classification techniques together. However, in previous studies segmentation and classification were performed individually. Furthermore, in previous studies, there is no comparison between the AlexNet and ResNet-34.

V. CONCLUSION

In this research, two pre-trained CNN architectures ResNet-34 and Alex-Net have been utilized for the classification of leukemia. It is founded that the ResNet-34 provides the best classification results as compared to Alex-Net. Pre-processing enhanced low-quality microscopic images and filters the noisy elements, resulting increase in performance prior to classification. Most of the previous researches employed a small, local dataset to evaluate their proposed algorithm. However, in this proposed framework, to maximize the number of images in the ALL-IDB dataset, used the data augmentation technique.

After the pre-processing step, the segmentation method is used to highlight the tumor area, and then classification is performed. A pre-trained network was used to create a computationally efficient detection system that uses learned topologies and parameters to produce competitive results. According to a statistical analysis of networks from the ALL-IDB dataset, the newly proposed technique has improved accuracy and is a sophisticated technique that could be beneficial in real-time medical applications.

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